香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Analysis Part!

ELEC 3200: System Modeling, Analysis and Control

Lecture 7: Laplace Transform

Yajing Shen (申亞京)

Office: 2450

eeyajing@ust.hk

Laplace Transform

A **signal** $x(t)$ is a real-valued function of time, i.e., $x(t)$: $\mathbb{R} \to \mathbb{R}$.

- **Laplace transform**

$$X(s) = \mathcal{L}\left[x(t)\right] = \int_{-\infty}^{\infty} x(t)e^{-st}dt.$$

- Unilateral signals: signals $x(t)$ with $x(t) = 0$ for all $t < 0$,

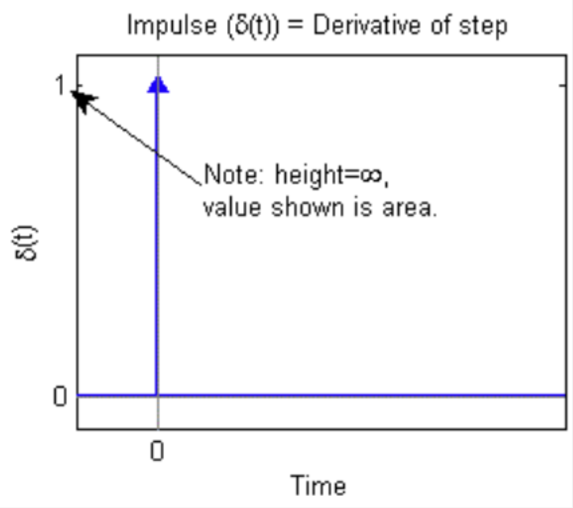
$$X(s) = \mathcal{L}\left[x(t)\right] = \int_{0^-}^{\infty} x(t)e^{-st}dt.$$

- **Region of convergence** (ROC): the set of complex number s that makes the integral meaningful.

Laplace Transform

▶ **Example** The Laplace transform of $\delta(t)$

$$\Delta(s) = \mathcal{L}[\delta(t)] = \int_{0^-}^{0^+} \delta(t) e^{-st} dt = 1.$$

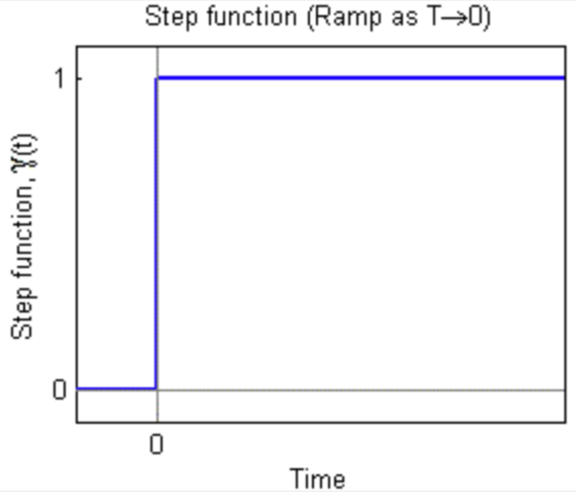


Unit impulse

▶ **Example** The Laplace transform of $\sigma(t)$

$$\Sigma(s) = \mathcal{L}[\sigma(t)] = \int_{0^-}^{\infty} e^{-st} dt = \begin{cases} \frac{1}{s}, & \mathrm{Re}\, s > 0; \\ \textbf{undefined}, & \mathrm{Re}\, s \le 0. \end{cases}$$

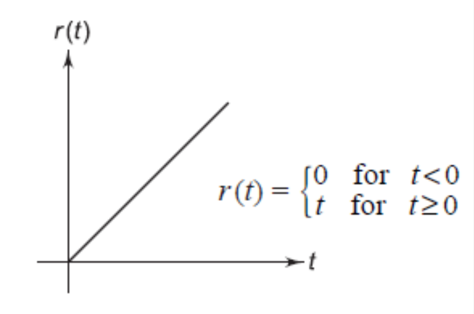
More common to say $\Sigma(s) = \mathcal{L}[\sigma(t)] = \frac{1}{s}$ with an ROC $\mathrm{Re}\, s > 0$.



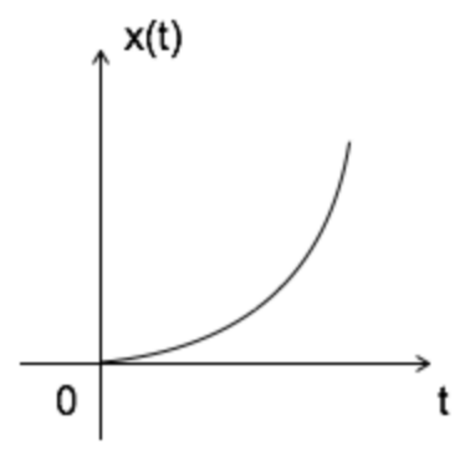
Unit step

Laplace Transform Table

Function name	$x(t)$	$X(s)$	ROC
Unit impulse	$\delta(t)$	1	$\mathbb{C}$
Unit step	$\sigma(t)$	$\frac{1}{s}$	$\mathrm{Re}\, s > 0$
Unit ramp	$t\sigma(t)$	$\frac{1}{s^2}$	$\mathrm{Re}\, s > 0$
Unit parabola	$\frac{t^2}{2}\sigma(t)$	$\frac{1}{s^3}$	$\mathrm{Re}\, s > 0$
nth-order ramp	$\frac{t^n}{n!}\sigma(t)$	$\frac{1}{s^{n+1}}$	$\mathrm{Re}\, s > 0$



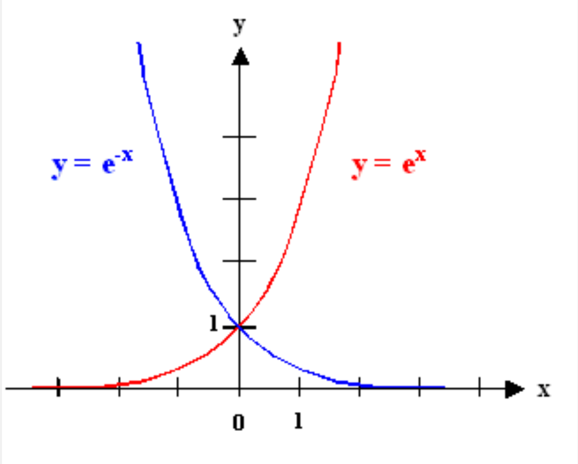
Unit ramp



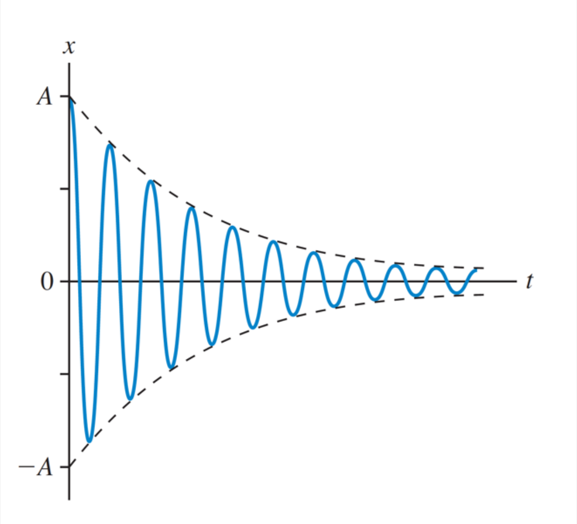
Unit parabola

Laplace Transform Table

Exponential	$e^{-\alpha t}\sigma(t)$	$\dfrac{1}{s+\alpha}$	$\mathrm{Re}\, s > -\alpha$
nth-order exponential	$\dfrac{t^n}{n!}e^{-\alpha t}\sigma(t)$	$\dfrac{1}{(s+\alpha)^{n+1}}$	$\mathrm{Re}\, s > -\alpha$
Sine	$(\sin \omega t)\sigma(t)$	$\dfrac{\omega}{s^2+\omega^2}$	$\mathrm{Re}\, s > 0$
Cosine	$(\cos \omega t)\sigma(t)$	$\dfrac{s}{s^2+\omega^2}$	$\mathrm{Re}\, s > 0$
Damped sine	$(e^{-\alpha t}\sin \omega t)\sigma(t)$	$\dfrac{\omega}{(s+\alpha)^2+\omega^2}$	$\mathrm{Re}\, s > -\alpha$
Damped cosine	$(e^{-\alpha t}\cos \omega t)\sigma(t)$	$\dfrac{s+\alpha}{(s+\alpha)^2+\omega^2}$	$\mathrm{Re}\, s > -\alpha$



Exponential



Damped sine

Properties of Laplace Transform

Number	Properties	Comments
1	$\mathcal{L}[x(t)] = X(s)$, $\mathcal{L}[y(t)] = Y(s)$	Notation
2	$\mathcal{L}[\alpha x(t) + \beta y(t)] = \alpha X(s) + \beta Y(s)$	Linearity
3	$\mathcal{L}[e^{-\alpha t}x(t)] = X(s+\alpha)$	Frequency shift
4	$\mathcal{L}[x(t-T)] = e^{-Ts}X(s)$	Time delay
5	$\mathcal{L}[x(\alpha t)] = \frac{1}{\alpha}X\left(\frac{s}{\alpha}\right)$	Scaling

Properties of Laplace Transform

Number	Properties	Comments
6	$\mathcal{L}\left[\dot{x}(t)\right] = sX(s)$	Derivative
7	$\mathcal{L}\left[x^{(n)}(t)\right] = s^n X(s)$	High-order derivative
8	$\mathcal{L}\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = \frac{X(s)}{s}$	Integration
9	$\mathcal{L}[x(t) * y(t)] = X(s)Y(s)$	Convolution

Properties of Laplace Transform

- **Derivative property** $\mathcal{L}\left[\dot{x}(t)\right] = sX(s)$

  - **Proof** Use **integration by parts**:

$$\mathcal{L}[\dot{x}(t)] = \int_{-\infty}^{\infty} \dot{x}(t)e^{-st}dt = \int_{-\infty}^{\infty} e^{-st}dx(t) = e^{-st}x(t)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x(t)de^{-st}.$$

The first term **vanishes** when s is in the ROC of $\mathcal{L}[x(t)]$. Hence

$$\mathcal{L}[\dot{x}(t)] = s\int_{-\infty}^{\infty} x(t)e^{-st}dt = s\mathcal{L}[x(t)]. \qquad \square$$

Properties of Laplace Transform

- **Integration property** $\mathcal{L}\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = \dfrac{X(s)}{s}$
  - **Proof** Let $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$. Then

$$\mathcal{L}[y(t)] = \int_{-\infty}^{\infty} y(t)e^{-st}dt = -\frac{1}{s}\int_{-\infty}^{\infty} y(t)de^{-st}.$$

Use **integration by parts**, then

$$\mathcal{L}[y(t)] = -\frac{1}{s}y(t)e^{-st}\Big|_{-\infty}^{\infty} + \frac{1}{s}\int_{-\infty}^{\infty} e^{-st}dy(t).$$

The first term **vanishes** when s is in the ROC of $\mathcal{L}[y(t)]$. Hence

$$\mathcal{L}[y(t)] = \frac{1}{s}\int_{-\infty}^{\infty} x(t)e^{-st}dt = \frac{1}{s}X(s).$$

$\square$

Properties of Laplace Transform

- **Convolution property** $\mathcal{L}[x(t) * y(t)] = X(s)Y(s)$
  - The convolution of two signals $x(t)$ and $y(t)$ is defined as

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(t-\tau)y(\tau)d\tau.$$

It is easy to see that

$$y(t) * x(t) = \int_{-\infty}^{\infty} y(t-\tau)x(\tau)d\tau = \int_{-\infty}^{\infty} y(\xi)x(t-\xi)d\xi = x(t) * y(t).$$

For unilateral signals $x(t)$ and $y(t)$,

$$x(t) * y(t) = \int_{0^-}^{t+0} x(t-\tau)y(\tau)d\tau.$$

Properties of Laplace Transform

- **Convolution property** $\mathcal{L}[x(t) * y(t)] = X(s)Y(s)$
  - **Proof** By definition and changing the order of the two integrations,

$$\mathcal{L}[x(t) * y(t)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(t-\tau)y(\tau)d\tau \right] e^{-st}dt$$
$$= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(t-\tau)e^{-s(t-\tau)}dt \right] y(\tau)e^{-s\tau}d\tau.$$

In the integral inside, set $t - \tau = \xi$. For fixed τ, when $t = \pm\infty$, it also holds $\xi = \pm\infty$. Then

$$\int_{-\infty}^{\infty} x(t-\tau)e^{-s(t-\tau)}dt = \int_{-\infty}^{\infty} x(\xi)e^{-s\xi}d\xi = X(s).$$

Therefore,

$$\mathcal{L}[x(t) * y(t)] = \int_{-\infty}^{\infty} X(s)y(\tau)e^{-s\tau}d\tau = X(s)Y(s).$$

Properties of Laplace Transform

- **Example** Let $x(t)$ be arbitrary and $y(t) = \delta(t)$.

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(t-\tau)\delta(\tau)d\tau = x(t),$$

$$\mathcal{L}[x(t) * \delta(t)] = X(s)\Delta(s) = X(s).$$

- **Example** Let $x(t)$ be arbitrary and $y(t) = \sigma(t)$.

$$x(t) * \sigma(t) = \int_{-\infty}^{\infty} x(\tau)\sigma(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)d\tau,$$

$$\mathcal{L}[x(t) * \sigma(t)] = X(s)\Sigma(s) = \frac{1}{s}X(s).$$

Inverse Laplace Transform

- If $X(s) = \mathcal{L}[x(t)]$, then $x(t) = \mathcal{L}^{-1}[X(s)]$.
- In this case, $x(t)$ is the **inverse Laplace transform** of $X(s)$.
- Let $\{\rho + j\omega : \omega \in \mathbb{R}\}$ be a vertical line in the ROC of $X(s)$. Then the formula for the inverse Laplace transform is

$$x(t) = \mathcal{L}^{-1}[X(s)] = \frac{1}{2\pi j}\int_{\rho-j\infty}^{\rho+j\infty} X(s)e^{st}ds.$$ **Useless!!**

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**

Let $G(s) = \dfrac{2}{s(s^2+2s+2)}$, find its inverse Laplace transform.

- First find the **partial fractional expansion** of $G(s)$

$$G(s) = \frac{2}{s(s+1-j)(s+1+j)} = \frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}.$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**
  - Then determine c_i of

$$G(s) = \frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}.$$

Multiply both sides by s, we get

$$sG(s) = c_1 + \frac{s}{s+1-j}c_2 + \frac{s}{s+1+j}c_3.$$

c_1 can be solved by

$$c_1 = \lim_{s\to 0} sG(s) = 1.$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**
  - Determine c_i of

$$G(s) = \frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}.$$

Similarly, c_2 and c_3 can be obtained by

$$c_2 = \lim_{s\to -1+j}(s+1-j)G(s) = \frac{1}{j(-1+j)}$$

$$c_3 = \lim_{s\to -1-j}(s+1+j)G(s) = \frac{1}{-j(-1-j)}.$$

c_3 can also be **simply** done as $c_3 = \bar{c}_2$.

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**
  - Another way to obtain c_i is by **comparing the coefficients**.

$$G(s) = \frac{2}{s(s+1-j)(s+1+j)} = \frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}.$$

Multiply each side by $s(s+1-j)(s+1+j)$, we get

$$\begin{aligned}
2 &= s(s+1-j)(s+1+j)\left(\frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}\right) \\
&= c_1(s+1-j)(s+1+j) + c_2 s(s+1+j) + c_3 s(s+1-j) \\
&= (c_1+c_2+c_3)s^2 + \left[2c_1 + (1+j)c_2 + (1-j)c_3\right] s + 2c_1.
\end{aligned}$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**
  - Comparing **the coefficients**, we get

$$\begin{aligned} c_1 + c_2 + c_3 &= 0 \\ 2c_1 + (1+j)c_2 + (1-j)c_3 &= 0 \\ 2c_1 &= 2. \end{aligned}$$

This gives

$$c_1 = 1, \quad c_2 = -\frac{1}{2} + \frac{1}{2}j, \quad c_3 = -\frac{1}{2} - \frac{1}{2}j,$$

which are the **same** as what we obtained using the first method.

Inverse Laplace Transform

- **Partial fractional expansion of a rational function**
  - Therefore

$$\begin{aligned}
\mathcal{L}^{-1}[G(s)] &= \mathcal{L}^{-1}[\frac{c_1}{s} + \frac{c_2}{s+1-j} + \frac{c_3}{s+1+j}] \\
&= \left[1 + \left(-\frac{1}{2} + \frac{1}{2}j\right) e^{(-1+j)t} + \left(-\frac{1}{2} - \frac{1}{2}j\right) e^{(-1-j)t}\right] \sigma(t) \\
&= \left[1 + 2e^{-t}\left(-\frac{1}{2}\cos t - \frac{1}{2}\sin t\right)\right] \sigma(t) \\
&= \left[1 - e^{-t}(\cos t + \sin t)\right] \sigma(t).
\end{aligned}$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**

Let $G(s) = \dfrac{1}{(s+1)^2 s^3}$, find its inverse Laplace transform.

- Let expand $G(s)$ as

$$G(s) = \frac{c_{11}}{s+1} + \frac{c_{12}}{(s+1)^2} + \frac{c_{21}}{s} + \frac{c_{22}}{s^2} + \frac{c_{23}}{s^3}.$$

Multiply both sides by $(s+1)^2$, we get

$$(s+1)^2 G(s) = (s+1)c_{11} + c_{12} + \frac{(s+1)^2}{s}c_{21} + \frac{(s+1)^2}{s^2}c_{22} + \frac{(s+1)^2}{s^3}c_{23}.$$

$$\Longrightarrow c_{12} = \lim_{s \to -1} (s+1)^2 G(s) = -1.$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**

$$(s+1)^2G(s) = (s+1)c_{11} + c_{12} + \frac{(s+1)^2}{s}c_{21} + \frac{(s+1)^2}{s^2}c_{22} + \frac{(s+1)^2}{s^3}c_{23}.$$

- Take the **derivative** of both sides, we get

$$\begin{aligned}
\frac{d}{ds}[(s+1)^2G(s)] &= \frac{d}{ds}\left[(s+1)c_{11} + c_{12} + \frac{(s+1)^2}{s}c_{21} + \frac{(s+1)^2}{s^2}c_{22} + \frac{(s+1)^2}{s^3}c_{23}\right] \\
&= c_{11} + 2(s+1)\left[\frac{1}{s}c_{21} + \frac{1}{s^2}c_{22} + \frac{1}{s^3}c_{23}\right] \\
&\quad + \frac{d}{ds}\left[\frac{1}{s}c_{21} + \frac{1}{s^2}c_{22} + \frac{1}{s^3}c_{23}\right](s+1)^2
\end{aligned}$$

$$\Longrightarrow c_{11} = \frac{1}{1!}\lim_{s\to -1}\frac{d}{ds}[(s+1)^2G(s)] = -3.$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**

$$G(s) = \frac{c_{11}}{s+1} + \frac{c_{12}}{(s+1)^2} + \frac{c_{21}}{s} + \frac{c_{22}}{s^2} + \frac{c_{23}}{s^3}.$$

  - Similarly, c_{21}, c_{22} and c_{23} can be obtained by

$$\begin{aligned} c_{21} &= \frac{1}{2!} \lim_{s \to 0} \frac{d^2}{ds^2} s^3 G(s) = 3 \\ c_{22} &= \frac{1}{1!} \lim_{s \to 0} \frac{d}{ds} s^3 G(s) = -2 \\ c_{23} &= \lim_{s \to 0} s^3 G(s) = 1. \end{aligned}$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**
  - Another way to find these coefficients is by **comparing the coefficients**.

$$G(s) = \frac{1}{(s+1)^2 s^3} = \frac{c_{11}}{s+1} + \frac{c_{12}}{(s+1)^2} + \frac{c_{21}}{s} + \frac{c_{22}}{s^2} + \frac{c_{23}}{s^3}$$

Multiply each side by $(s+1)^2 s^3$, we get

$$\begin{aligned} 1 &= c_{11}(s+1)s^3 + c_{12}s^3 + c_{21}(s+1)^2 s^2 + c_{22}(s+1)^2 s + c_{23}(s+1)^2 \\ &= (c_{11}+c_{21})s^4 + (c_{11}+c_{12}+2c_{21}+c_{22})s^3 + (c_{21}+2c_{22}+c_{23})s^2 + (c_{22}+2c_{23})s + c_{23} \end{aligned}$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**
  - By comparing **the coefficients**, obtain a set of linear equations

$$\begin{aligned} c_{11} + c_{21} &= 0 \\ c_{11} + c_{12} + 2c_{21} + c_{22} &= 0 \\ c_{21} + 2c_{22} + c_{23} &= 0 \\ c_{22} + 2c_{23} &= 0 \\ c_{23} &= 1. \end{aligned}$$

  - The solution is

$$c_{11} = -3, \quad c_{12} = -1, \quad c_{21} = 3, \quad c_{22} = -2, \quad c_{23} = 1.$$

Inverse Laplace Transform

- **Partial fractional expansion of a rational function (multiple poles)**
  - We obtain
$$G(s) = \frac{-3}{s+1} + \frac{-1}{(s+1)^2} + \frac{3}{s} + \frac{-2}{s^2} + \frac{1}{s^3}.$$
  - Finally, the inverse Laplace transform can then be obtained from **Laplace transform table** and the **linearity property**:
$$\mathcal{L}^{-1}[G(s)] = \left[-(3+t)e^{-t} + 3 - 2t + \frac{1}{2}t^2\right]\sigma(t).$$

Review & Exercise

- Know the concept of Laplace transform
- Know the basic Laplace transform table
- Know the basic properties of Laplace transform
- Know the methods of Inverse Laplace transform
- Be familiar with the relevant mathematic skills

Appendix

More about Laplace transform

The Laplace transform is an integral transform perhaps second only to the Fourier transform in its utility in solving physical problems.

The Laplace transform is particularly useful in solving linear ordinary differential equations such as those arising in the analysis of electronic circuits.

The most significant advantage is that it turns integral equations and differential equations to polynomial equations, which are much easier to solve. Once solved, use of the inverse Laplace transform reverts to the original domain.

The key idea behind the Laplace transform is to transform a time-domain representation of a system into a complex frequency-domain representation. This transformation simplifies the analysis of linear time-invariant systems and differential equations.

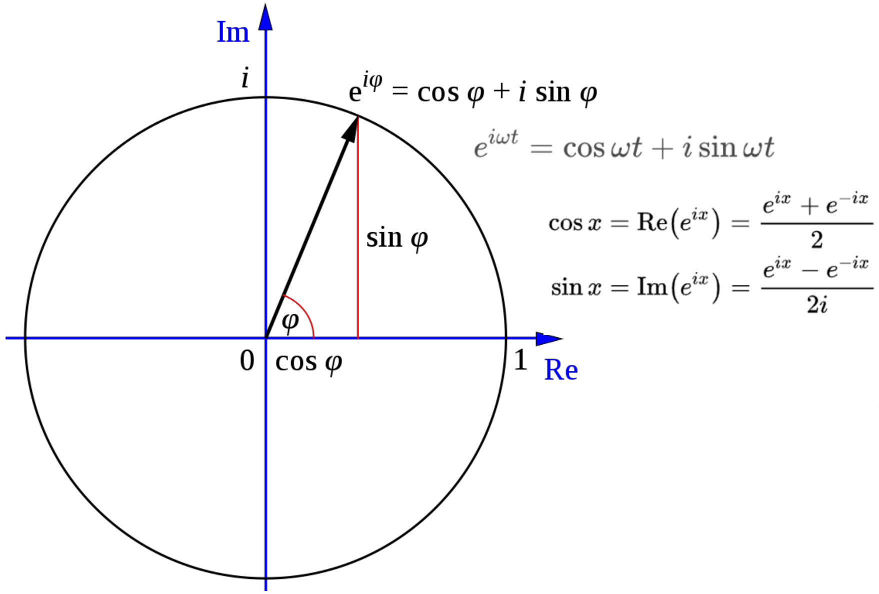
$$m\frac{d^2x}{dt} + c\frac{dx}{dt} + kx = 0 \longrightarrow mp^2 + cp + k = 0$$

Pierre-Simon, marquis de Laplace

1749-1827

More about Laplace transform

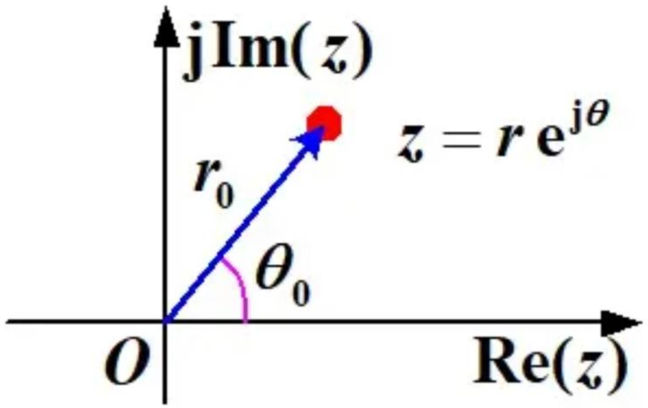
Euler's Formula



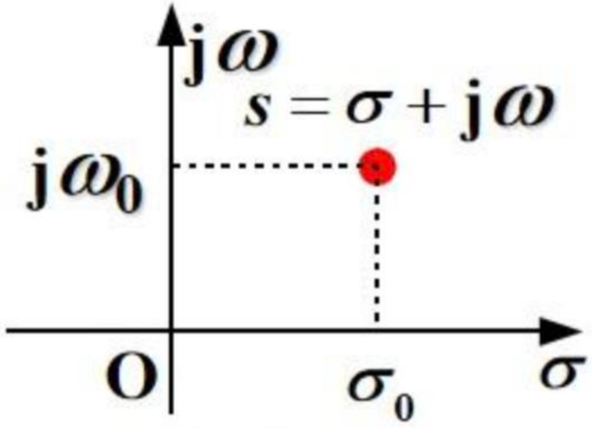
$$e^{ix} = \cos x + i\sin x$$

Euler's formula is ubiquitous in mathematics, physics, chemistry, and engineering. The physicist Richard Feynman called the equation "our jewel" and "the most remarkable formula in mathematics".

z-plane



s-plane



$$z = e^{sT}$$

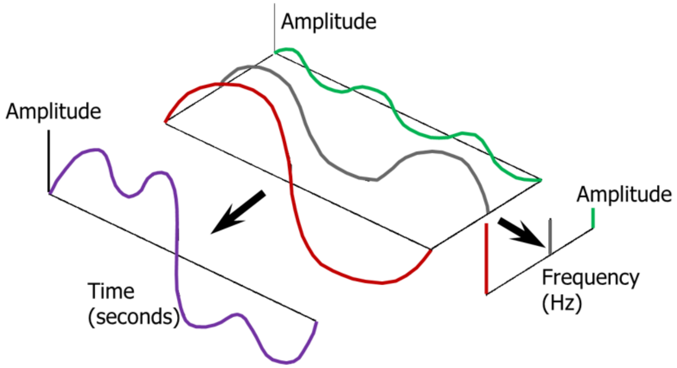
$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} \cdot e^{j\omega T} = e^{\sigma T} \angle \omega T \qquad s = \sigma + j\omega$$

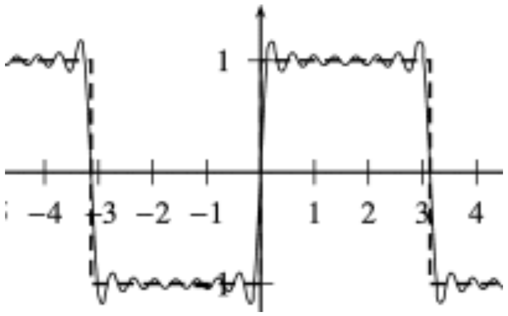
$$\begin{cases} R = |z| = e^{\sigma T} \\ \theta = \angle z = \omega T \end{cases}$$

The real part of s

The imaginary part of s

More about Laplace transform



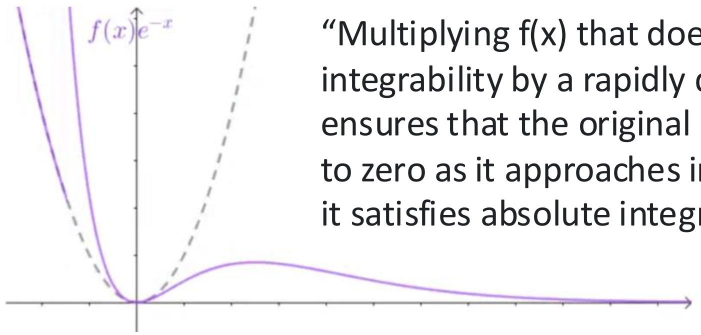


In order for a function f (x) to be expanded properly, it must satisfy the following Dirichlet conditions: A piecewise function f (x) must be periodic with at most a finite number of discontinuities, and/or a finite number of minima or maxima within one period.

$$F(\omega)=\int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}d\omega$$

Fourier transform



“Multiplying f(x) that does not satisfy absolute integrability by a rapidly decaying function ensures that the original function also decays to zero as it approaches infinity. Consequently, it satisfies absolute integrability."

$$\lim_{x\to+\infty} f(x)e^{-\sigma x}=0, \sigma\in R$$

In order to ensure that $e^{-\sigma x}$ remains a decaying function, we restrict the domain to the positive half-axis. This allows us to perform the Fourier transform, resulting in:

$$F(\omega)=\int_0^{+\infty} f(t)e^{-\sigma t}e^{-i\omega t}dt=\int_0^{+\infty} f(t)e^{-(\sigma+i\omega)t}dt$$

Define: $s=\sigma+i\omega$

Then, we have:

$$F(s)=\int_0^{+\infty} f(t)e^{-st}dt$$

Laplace transform

More about Laplace transform

Story

In 1807, **Fourier** submitted a paper to the French Academy of Sciences, using sine curves to describe temperature distribution. The paper contained a controversial idea at the time: any continuous periodic signal could be represented by an appropriate combination of sine curves. Fourier did not provide a rigorous mathematical proof.

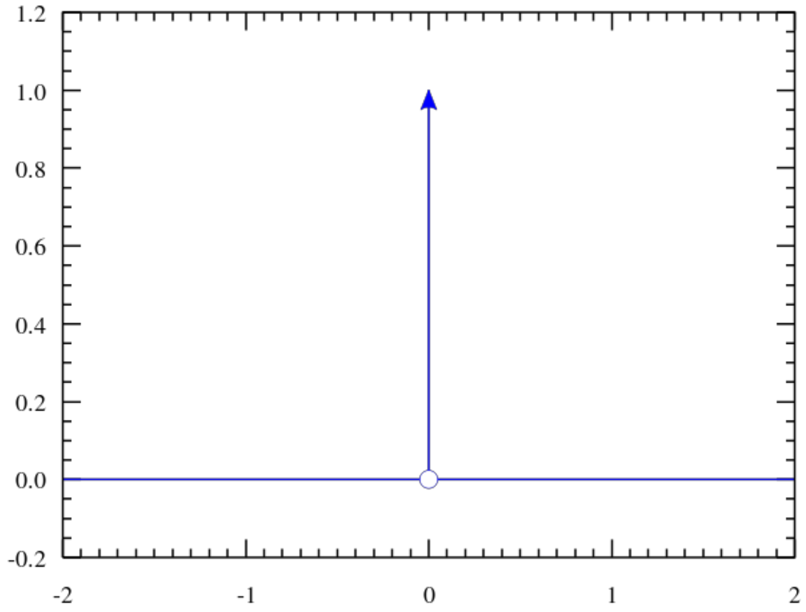
Among the reviewers of this paper were the historically famous mathematicians Lagrange and Laplace. When **Laplace** and other reviewers voted to approve and publish the paper, **Lagrange** strongly opposed, believing that Fourier's method could not represent signals with corners. The French Academy of Sciences yielded to Lagrange's prestige and rejected Fourier's work.

In 1822, Fourier transforms were published along with his work "Analytical Theory of Heat," but it was already 15 years later.

In 1829, **Dirichlet**, by deriving its scope of application, improved the Fourier transform.

More about Dirac delta function

$$\delta(x) \simeq \begin{cases} +\infty, & x = 0 \\ 0, & x \neq 0 \end{cases}$$



It is also constrained to satisfy the identity:

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1.$$

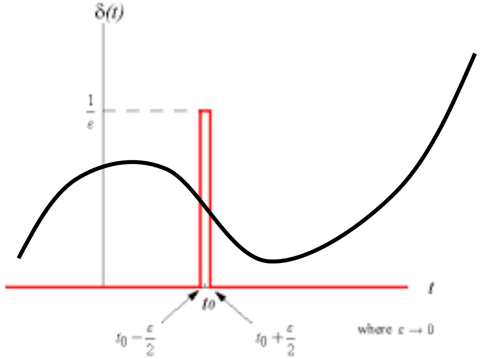
Another equivalent definition of the Dirac delta function:

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1, \quad \int_{-\infty}^{\infty} \delta(x) g(x) dx = g(0)$$

The Dirac delta function can be rigorously defined either as a distribution or as a measure:

$$f(t) = f(t) * \delta(t)$$

The convolution of any function with the impulse function is equal to the function itself.



$$\int_{-\infty}^{\infty} f(t)\, \delta(t - T)\, dt = f(T)$$

This is sometimes referred to as the sifting property or the sampling property. The delta function is said to "sift out" the value of f(t) at t = T.

Calculus skills - the integration by parts

$$\int u\,dv = uv - \int v\,du$$

Example1:

In order to calculate

$$I = \int x\cos(x)\,dx\,,$$

let:

$$\begin{aligned} u &= x & &\Rightarrow\ du = dx \\ dv &= \cos(x)\,dx \Rightarrow & v &= \int \cos(x)\,dx = \sin(x) \end{aligned}$$

then:

$$\begin{aligned} \int x\cos(x)\,dx &= \int u\,dv \\ &= u\cdot v - \int v\,du \\ &= x\sin(x) - \int \sin(x)\,dx \\ &= x\sin(x) + \cos(x) + C, \end{aligned}$$

where C is a constant of integration.

Example2:

$$I = \int e^x\cos(x)\,dx.$$

Here, integration by parts is performed twice. First let

$$\begin{aligned} u &= \cos(x) \Rightarrow du = -\sin(x)\,dx \\ dv &= e^x\,dx \ \Rightarrow \ v = \int e^x\,dx = e^x \end{aligned}$$

then:

$$\int e^x\cos(x)\,dx = e^x\cos(x) + \int e^x\sin(x)\,dx.$$

Now, to evaluate the remaining integral, we use integration by parts again, with:

$$\begin{aligned} u &= \sin(x) \Rightarrow du = \cos(x)\,dx \\ dv &= e^x\,dx \ \Rightarrow \ v = \int e^x\,dx = e^x. \end{aligned}$$

Then:

$$\int e^x\sin(x)\,dx = e^x\sin(x) - \int e^x\cos(x)\,dx.$$

Putting these together,

$$\int e^x\cos(x)\,dx = e^x\cos(x) + e^x\sin(x) - \int e^x\cos(x)\,dx.$$

Calculus skills - L'Hôpital's rule

L'Hôpital's rule states that for functions *f* and *g* which are differentiable on an open interval *I* except possibly at a point *c* contained in *I*, if:

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0 \text{ or } \pm\infty$$

$$g'(x) \neq 0 \text{ for all } x \text{ in } I \text{ with } x \neq c,$$

$$\lim_{x\to c} \frac{f'(x)}{g'(x)} \text{ exists}$$

$\Rightarrow$

$$\boxed{\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}}$$

The differentiation of the numerator and denominator often simplifies the quotient or converts it to a limit that can be evaluated directly.

Example1:

$$\lim_{x\to 0} \frac{e^x - 1}{x^2 + x} = \lim_{x\to 0} \frac{\frac{d}{dx}(e^x - 1)}{\frac{d}{dx}(x^2 + x)} = \lim_{x\to 0} \frac{e^x}{2x+1} = 1.$$

Example2:

$$\lim_{x\to\infty} x^n \cdot e^{-x} = \lim_{x\to\infty} \frac{x^n}{e^x} = \lim_{x\to\infty} \frac{nx^{n-1}}{e^x} = n \cdot \lim_{x\to\infty} \frac{x^{n-1}}{e^x}$$

Repeatedly apply L'Hôpital's rule until the exponent is zero (if n is an integer) or negative (if n is fractional) to conclude that the limit is zero.

Example3:

$$\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\to 0^+} -x = 0.$$

More about convolution

$$(f * g)(t) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau$$

To convolve a kernel with an input signal:
flip the signal, move to the desired time,
and accumulate every interaction with the kernel

$$f * g = g * f$$

$$\int (f * g) = \int f \cdot \int g \qquad \int (x \cdot x) \neq \int x \cdot \int x$$

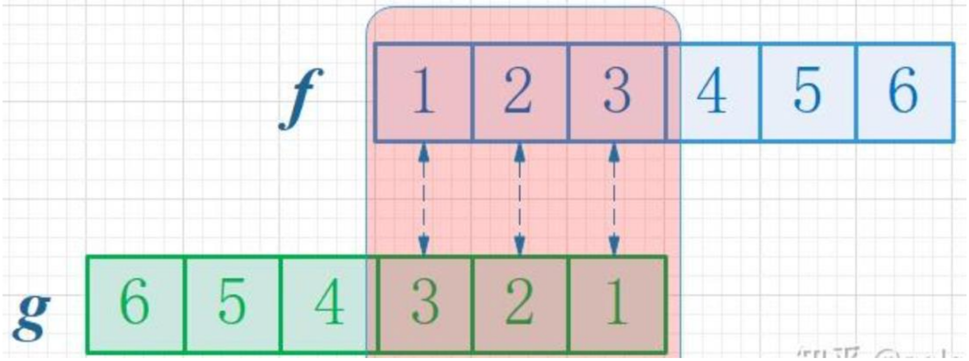
For example:

- $\int (x \cdot x) = \int x^2 = \frac{1}{3}x^3$
- $\int x \cdot \int x = \frac{1}{2}x^2 \cdot \frac{1}{2}x^2 = \frac{1}{4}x^4$

Applications (e.g.)

1.Signal Analysis: An input signal f(t), when passed through a linear system (whose characteristics can be described by the unit impulse response function g(t)). The output signal can be obtained through convolution operations.

2.Image Processing: Input an image f(x, y), and after convolution with a specifically designed kernel g(x, y), the output image will exhibit various effects such as blurring, edge enhancement, and more.

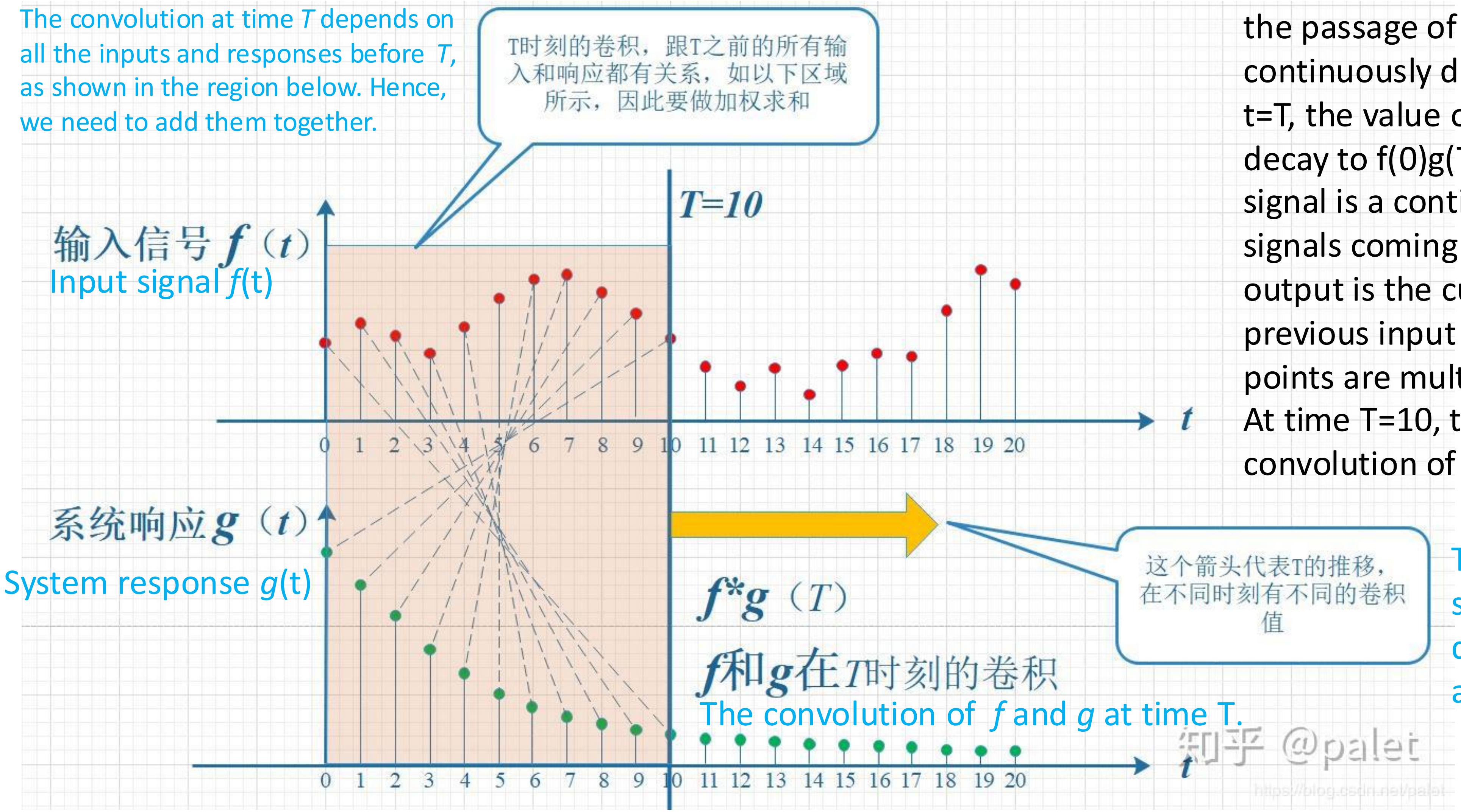


$$f(1)g(3) + f(2)g(2) + f(3)g(1)$$

is $\quad (f * g)(4) = \sum_{m=1}^{3} f(4-m)g(m)$

More about convolution

The convolution at time *T* depends on all the inputs and responses before *T*, as shown in the region below. Hence, we need to add them together.



If there is an input at time t=0, then, with the passage of time, this input will continuously decay. In other words, at time t=T, the value of the input f(0) at t=0 will decay to f(0)g(T). Considering that the signal is a continuous input, with new signals coming in at each moment, the final output is the cumulative effect of all previous input signals. These corresponding points are multiplied and then accumulated. At time T=10, the output signal value is the convolution of functions f and g at T=10.

This arrow represents the shift of T, resulting in different convolution values at different instants.

https://www.zhihu.com/question/22298352